

1_10xMouse_PBMC_Stellarscope_pseudobulk

August 31, 2026

```
[2]: library(Seurat)
library(ggplot2)
library(pheatmap)
library(dplyr)
library(RColorBrewer)
library(reshape2)
library(clustree)
getwd()
dir.create("figures_10xMouse_PBMC")
dir.create("data_10xMouse_PBMC")
dataset_id <- "10xMouse_PBMC"
sample <- "10k"
mode <- "pseudobulk"
RESULTS_PATH_Stellarscope <- "/mnt/TEresults2/snakemake_results"

colorPBMC <- "#81B29A"
```

```
'/mnt/TEdata/TEbenchmarking/jupyterNotebooks/realDatasets'
```

```
Warning message in dir.create("figures_10xMouse_PBMC"):
```

```
"'figures_10xMouse_PBMC' already exists"
```

```
Warning message in dir.create("data_10xMouse_PBMC"):
```

```
"'data_10xMouse_PBMC' already exists"
```

```
[13]: # celltype annotation

celltypes <- read.table(paste0("data_", dataset_id, "/celltype_annotation.tsv"),
  sep = "\t")
colnames(celltypes) <- c("barcode", "celltype")
rownames(celltypes) <- celltypes$barcode

PBMCcelltypeColors <- c("B_cells"="#6D5A5D",
  "Dendritic Cells"="#C4B3AB", "#E78063", "#c492a7",
  "Monocytes"="#81B29A",
  "Platelets"="#e78063",
  "T_cells"="#84B6D6",
  "Natural Killers"="#F2CC8F",
  "Neutrophils"="#c492a7",
```

```
"Unknown" = "#3D405B")
```

```
[32]: path <- paste0("/mnt/TEresults2/snake_make_results/results/stellarscope_out/",  
  ↪ dataset_id, "/", sample, "/", mode, "/")  
STAR_path <- paste0("/mnt/TEresults/snake_make_results/results/STARoutdir/  
  ↪", dataset_id, "/", sample, "/best_Solo.out/Gene")  
  
filteredBarcodes <- read.table(paste0(STAR_path, "/filtered/barcodes.tsv"))$V1  
  ↪ # read barcodes selected by STARsolo  
# remove cells annotated as multiplets by 10x  
mouse_assignment <- read.csv("data_10xMouse_PBMC/  
  ↪ SC3_v3_NextGem_DI_CellPlex_Mouse_PBMC_10K_Multiplex_multiplexing_analysis_assignment_confid  
  ↪ csv")  
mouse_assignment$Barcodes <- sapply(strsplit(mouse_assignment$Barcodes,  
  ↪ split="-"), "[", 1)  
multiplets <-  
  ↪ mouse_assignment$Barcodes[mouse_assignment$Assignment=="Multiplet"]  
filteredBarcodes <- setdiff(filteredBarcodes, multiplets)  
  
TEmatrix <- Seurat::ReadMtx(mtx = paste0(path, sample, "_", mode, "-TE_counts.  
  ↪ mtx"),  
  cells = paste0(path, sample, "_", mode,  
  ↪ "-barcodes.tsv"),  
  features = paste0(path, sample, "_", mode,  
  ↪ "-features.tsv"),  
  feature.column = 1) # read matrix  
TEmatrix <-  
  ↪ TEmatrix[setdiff(rownames(TEmatrix), "__no_feature"), filteredBarcodes]  
  
nCells <- ncol(TEmatrix)  
thrMinCells <- round(nCells * 0.05)  
  
# create Seurat object  
objTE_stellarscope <- Seurat::CreateSeuratObject(TEmatrix, project = "PBMC",  
  min.cells = thrMinCells, min.features = 20)
```

Warning message:

"Feature names cannot have underscores ('_'), replacing with dashes ('-')"

```
[33]: objTE_stellarscope
```

An object of class Seurat

4165 features across 4787 samples within 1 assay

Active assay: RNA (4165 features, 0 variable features)

1 layer present: counts

```
[34]: annotation_stellarscope <- read.table("annotation/annotation_stellarscope.tsv")
      # generated with annotation_scripts/create_annotations_mouse.Rmd
      head(annotation_stellarscope)
```

		chr	start	end	sw_score	strand	locusID	family	subfam
		<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>	<chr>
A data.frame: 6 × 10	1	chr1	3000001	3002128	12955	-	L1_Mus3_dup24	L1	L1-Mu
	2	chr1	3003153	3003994	1216	-	L1Md_F_dup2	L1	L1Md-
	3	chr1	3003994	3004054	234	-	L1_Mus3_dup25	L1	L1-Mu
	4	chr1	3004041	3004206	3685	+	L1_Rod_dup1	L1	L1-Ro
	5	chr1	3004271	3005001	3685	+	L1_Rod_dup2	L1	L1-Ro
	6	chr1	3005002	3005439	1280	+	L1_Rod_dup3	L1	L1-Ro

```
[35]: # Remove some classes of TEs

classesToExclude <- c("Other", "Satellite", "Unknown", "RNA")

stellarscopeTEs <- Features(objTE_stellarscope)

TEsToKeep <- annotation_stellarscope[! annotation_stellarscope$class %in%
  # classesToExclude, ]$stellarscopeID

length(setdiff(stellarscopeTEs, TEsToKeep))/ length(stellarscopeTEs) * 100 #
  # percentage removed

objTE_stellarscope <- objTE_stellarscope[intersect(stellarscopeTEs, TEsToKeep),]
```

1.00840336134454

```
[36]: feature_metadata <- annotation_stellarscope[match(Features(objTE_stellarscope),
  # annotation_stellarscope$stellarscopeID),]

      head(feature_metadata)
```

		chr	start	end	sw_score	strand	locusID	family
		<chr>	<int>	<int>	<int>	<chr>	<chr>	<chr>
A data.frame: 6 × 10	2345	chr1	4897602	4897737	1203	+	B1_Mus2_dup51	Alu
	2642	chr1	5086609	5086792	1311	+	B2_Mm2_dup46	B2
	2674	chr1	5102798	5102943	1219	-	B1_Mus2_dup54	Alu
	2675	chr1	5102985	5103159	285	-	MIRb_dup22	MIR
	13318	chr1	13158124	13158508	3022	-	L1Md_A_dup120	L1
	13485	chr1	13324289	13324478	1645	+	B2_Mm1t_dup99	B2

```
[37]: options(repr.plot.width=7, repr.plot.height=5)

objTE_stellarscope@meta.data$nCount_TE <- objTE_stellarscope@meta.
  # data$nCount_RNA
```

```

objTE_stellarscope@meta.data$nFeature_TE <- objTE_stellarscope@meta.
  ↪data$nFeature_RNA
# Visualize QC metrics as a violin plot
VlnPlot(objTE_stellarscope, features = c("nCount_TE"), ncol = 1,
  cols = colorPBM, pt.size = 0) + theme(text=element_text(size=17))

ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_stellarscope_",mode,".
  ↪png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nCountTE_violin_stellarscope_",mode,".
  ↪pdf"), device='pdf')

VlnPlot(objTE_stellarscope, features = c("nFeature_TE"), ncol = 1,
  cols = colorPBM, pt.size = 0) + theme(text=element_text(size=17))
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_stellarscope_",mode,".
  ↪png"), device='png',dpi=600)
ggsave(paste0("figures_",dataset_id,"/nFeatureTE_violin_stellarscope_",mode,".
  ↪pdf"), device='pdf')

```

Warning message:

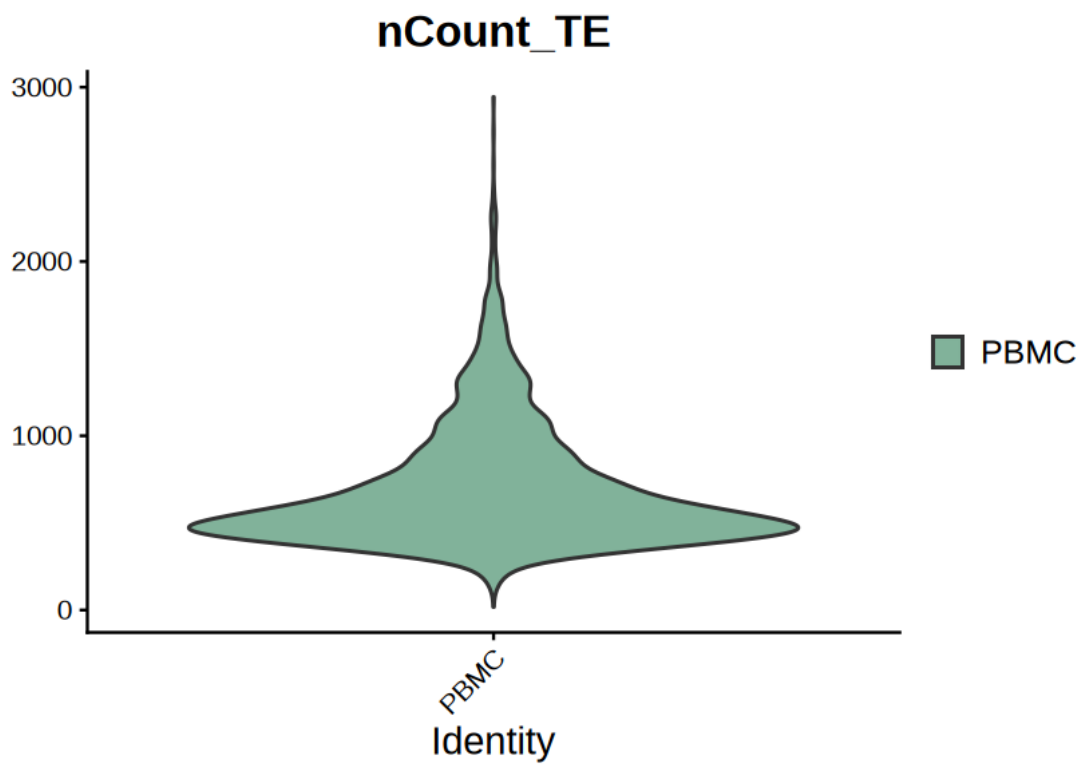
"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."

Saving 6.67 x 6.67 in image

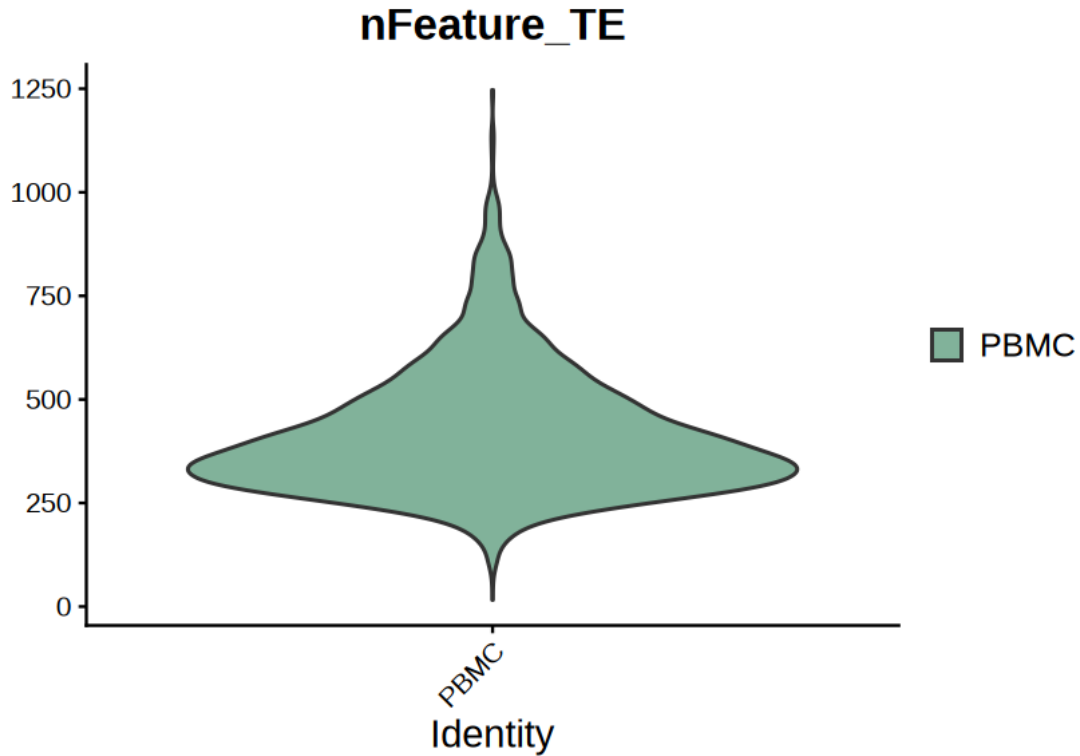
Saving 6.67 x 6.67 in image

Warning message:

"Default search for "data" layer in "RNA" assay yielded no results; utilizing
"counts" layer instead."



Saving 6.67 x 6.67 in image
Saving 6.67 x 6.67 in image



```
[38]: objTE_stellarscope <- JoinLayers(objTE_stellarscope)

objTE_stellarscope <- NormalizeData(objTE_stellarscope, normalization.method = "LogNormalize",
  ↪ scale.factor = 10000)

objTE_stellarscope <- FindVariableFeatures(objTE_stellarscope, selection.method = "vst",
  ↪ nfeatures = 4000)

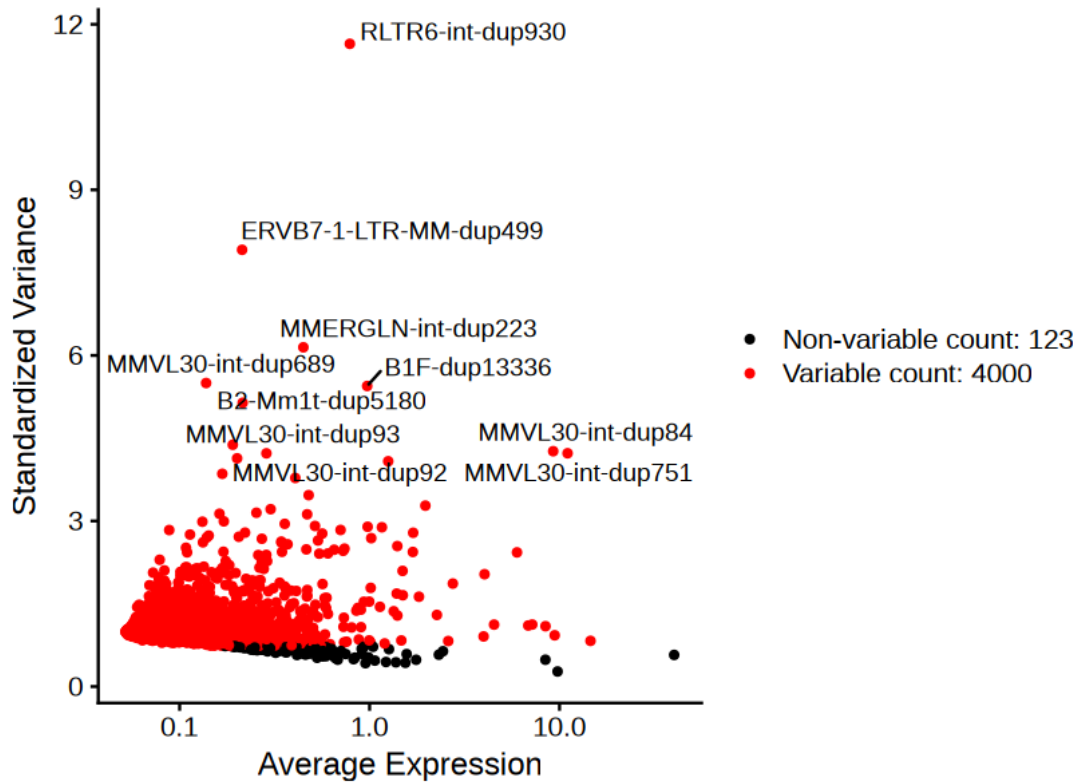
# Identify the 10 most highly variable genes
top10 <- head(VariableFeatures(objTE_stellarscope), 10)

# plot variable features with and without labels
plot1 <- VariableFeaturePlot(objTE_stellarscope)
plot2 <- LabelPoints(plot = plot1, points = top10, repel = TRUE)
plot2
```

Normalizing layer: counts

Finding variable features for layer counts

When using repel, set xnudge and ynudge to 0 for optimal results



```
[39]: gc()
all.genes <- rownames(objTE_stellarscope)
objTE_stellarscope <- ScaleData(objTE_stellarscope) # on hugs
```

		used	(Mb)	gc trigger	(Mb)	max used	(Mb)
A matrix: 2 × 6 of type dbl	Ncells	15310644	817.7	43535568	2325.1	36875254	1969.4
	Vcells	106882402	815.5	279271860	2130.7	279271860	2130.7

Centering and scaling data matrix

```
[40]: objTE_stellarscope <- RunPCA(objTE_stellarscope, features = □
↪VariableFeatures(object = objTE_stellarscope))

DimPlot(objTE_stellarscope, reduction = "pca") + NoLegend()

ElbowPlot(objTE_stellarscope)
```

PC_ 1

Positive: RSINE1-dup110789, RSINE1-dup7030, B1F2-dup8771, RMER5-dup4769,
MLT1A1-dup1566, L2c-dup11353, B1-Mur1-dup2948, RMER20B-dup1279, PB1D9-dup28225,
B3-dup44048

RSINE1-dup6708, L1-Mus3-dup16098, B1-Mus2-dup69004, B2-Mm2-dup1895,

RSINE1-dup6707, B1-Mur2-dup8227, LTR84a-dup93, Tigger14a-dup81, B1F-dup2259, MLT1B-dup8769

B2-Mm1a-dup4264, RLTR6-Mm-dup14, B1F-dup16609, Lx8-dup32585, Lx-dup23147, ID-B1-dup42350, MMVL30-int-dup751, L1Md-T-dup16708, MTD-dup36599, RLTR20C1-MM-dup5065

Negative: RLTR4-MM-int-dup288, B1-Mus1-dup50752, ERVB4-1C-LTR-Mm-dup282, RLTR6-Mm-dup145, RLTR4-MM-int-dup287, L1Md-F3-dup14136, RMER19B2-dup2236, PB1D9-dup18957, B1-Mur3-dup16020, B1-Mus1-dup50780

MER131-dup95, RLTR4-Mm-dup171, L1-Mus3-dup46173, MER20B-dup573, L1M2-dup15802, B1-Mus1-dup50755, B2-Mm1a-dup11535, MT2B1-dup2418, Lx9-dup46147, B2-Mm2-dup7180

Lx8b-dup1979, Zaphod2-dup7, B3-dup119403, PB1-dup9394, L1Md-F2-dup62953, L1-Mus3-dup42684, B2-Mm2-dup71426, B1-Mus1-dup94686, B1-Mus1-dup50766, B3A-dup62438

PC_ 2

Positive: L1Md-F3-dup14136, B1-Mur3-dup16020, PB1D9-dup18957, MER20B-dup573, L1M2-dup15802, B2-Mm1a-dup11535, B1-Mus1-dup50752, B1-Mus1-dup50780, L1-Mus3-dup42684, B1-Mus1-dup50755

MMVL30-int-dup751, MMVL30-int-dup84, Lx9-dup46147, B3A-dup62438, MMVL30-int-dup261, MMVL30-int-dup87, Lx3B-dup5224, B4-dup43291, ID4-dup16091, L1Md-F2-dup62443

B1-Mm-dup27896, ID-B1-dup95799, B1-Mus1-dup62785, L1-Mus3-dup46173, Lx7-dup27511, B1-Mus1-dup50766, B1-Mm-dup27897, L1ME2-dup1961, ID-B1-dup95802, MMVL30-int-dup885

Negative: Lx3B-dup2078, RLTR4-Mm-dup52, RLTR4-MM-int-dup130, RLTR4-Mm-dup171, ID-B1-dup44259, RLTR4-Mm-dup51, RLTR4-MM-int-dup288, B1-Mus2-dup27269, B1-Mm-dup34748, RLTR4-MM-int-dup287

RLTR53-Mm-dup209, B3-dup136807, L1Md-F2-dup23486, ID-B1-dup101661, PB1D11-dup5363, RSINE1-dup46907, ORR1C2-dup9583, L1Md-F2-dup57836, L1VL2-dup759, MIR-dup32006

B2-Mm2-dup86522, RLTR4-MM-int-dup306, RLTR4-MM-int-dup222, B3-dup99286, Lx2A-dup1173, MYSERV6-int-dup2636, B1-Mur2-dup27171, RLTR11A-dup2406, B1-Mus2-dup70692, L1-Mm-dup2361

PC_ 3

Positive: L1Md-F3-dup15875, ORR1D1-dup7350, Charlie8-dup493, ORR1F-dup8737, B1F-dup35309, L1MB5-dup1314, B1-Mus2-dup52936, B1F1-dup10203, ORR1D2-dup6027, B3A-dup34745

L1-Mur3-dup933, MLT1A0-dup8787, B1-Mus1-dup80571, L1-Mus3-dup1683, B2-Mm1t-dup657, B3A-dup2356, L1MD-dup6120, Lx-dup812, B1-Mur4-dup34716, B1-Mus1-dup80573

RSINE1-dup43494, B2-Mm1a-dup15134, B4-dup52848, B1-Mus1-dup61643, B4-dup1835, B1-Mus1-dup80569, RLTR9E-dup1210, B2-Mm2-dup35748, RSINE1-dup98867, MIRc-dup1740

Negative: MLT1A1-dup1566, MLT1B-dup8769, RSINE1-dup6708, RSINE1-dup6707, LTR84a-dup93, B2-Mm1a-dup4264, L1-Mus3-dup16098, B1F-dup13336, B1-Mm-dup14909, RMER15-dup5271

Tigger14a-dup81, ID-B1-dup33875, Lx7-dup13295, B3-dup139225, L1-Mus3-dup16097, RLTR6B-Mm-dup611, L1MA5-dup4632, L1-Mus4-dup4115, RLTR6-Mm-

dup14, B1-Mur1-dup2948

MMVL30-int-dup92, L2a-dup12848, RLTR6B-Mm-dup47, RLTR21-dup3247,
MER58A-dup3447, L1Md-F2-dup5378, RLTR6-int-dup930, RLTR6-Mm-dup166,
B1-Mus1-dup28222, L1Md-F3-dup5416

PC_ 4

Positive: ORR1D1-dup7350, L1Md-F3-dup15875, L1MB5-dup1314, B2-Mm2-dup82893,
RSINE1-dup43494, RLTR9D2-dup89, MIR-dup36062, MTB-dup1832, RLTR9E-dup1210, MTC-
dup7908

B1F1-dup10203, RSINE1-dup29247, B2-Mm1t-dup657, ORR1D2-dup6027,
Charlie4z-dup46, B1-Mus2-dup3098, B2-Mm2-dup14374, LTRIS2-dup151, Lx9-dup18072,
B1-Mus1-dup10288

Lx5b-dup6947, B1-Mur3-dup8989, B2-Mm1a-dup5375, B1F-dup15855,
B1-Mus1-dup61643, ORR1F-dup8737, B1-Mm-dup9547, B1-Mus2-dup29186,
RMER12C-dup1049, MTC-dup12054

Negative: L1-Mus3-dup46173, L1Md-F2-dup46200, Lx9-dup46147, L1-Mus1-dup947,
B2-Mm2-dup767, L1MEf-dup1497, RLTR44-int-dup505, B1-Mus1-dup50752,
RSINE1-dup60668, RLTR44-int-dup512

Lx-dup7783, L1Md-F2-dup5378, MER131-dup95, L1-Mus3-dup7164,
B1-Mus1-dup50780, RLTR44-int-dup502, MMVL30-int-dup685, MMVL30-int-dup84,
L1-Mus1-dup946, Lx8b-dup1979

Lx9-dup30117, L1Md-F3-dup14136, B3-dup74716, B4A-dup22917,
L1-Mus1-dup19220, L1Md-A-dup1448, L1Md-F2-dup62443, B4-dup5419, RLTR44-int-
dup507, MuLV-int-dup62

PC_ 5

Positive: MMVL30-int-dup685, B3-dup103051, ERVB7-1-LTR-MM-dup499, RLTR6-Mm-
dup145, B1-Mus2-dup63347, B3-dup136807, B2-Mm2-dup61191, Lx2B-dup1342, RLTR6-Mm-
dup166, MMTV-int-dup4

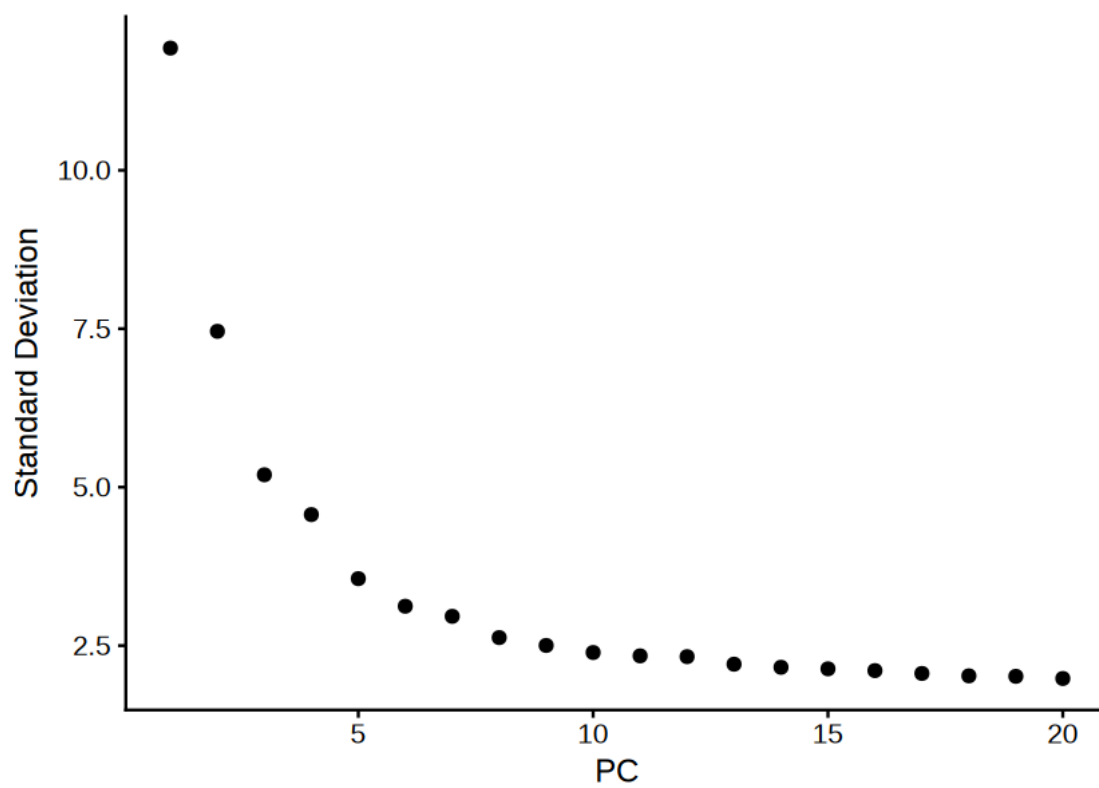
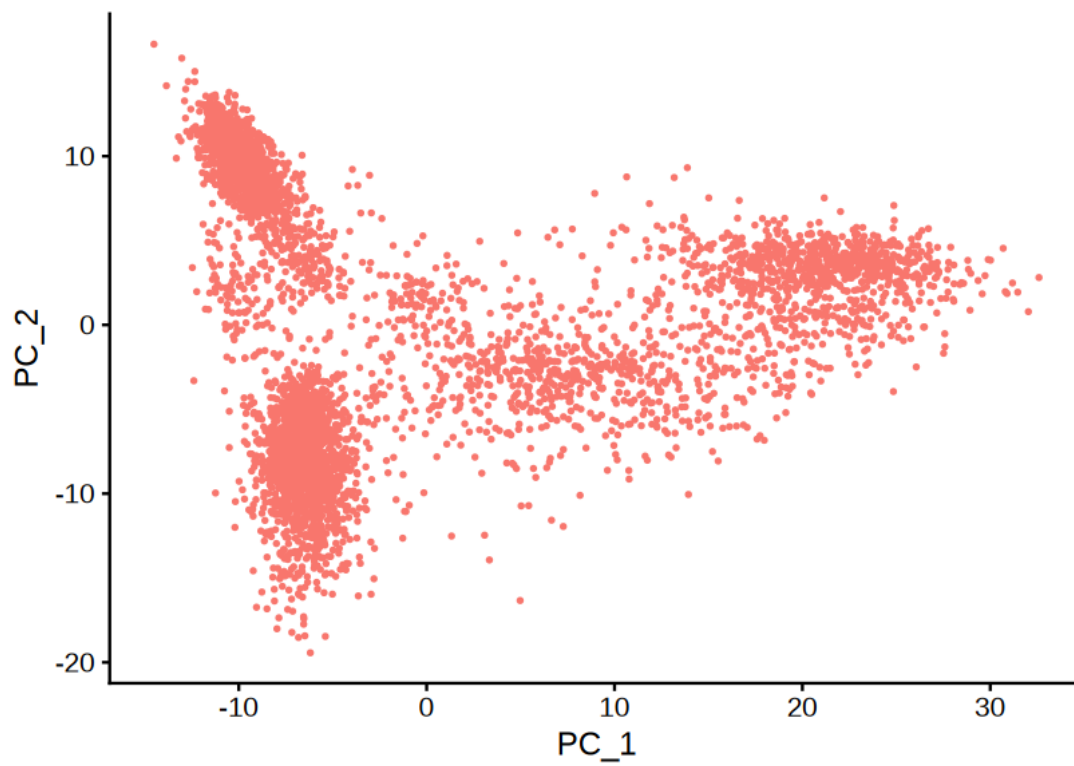
L1-Mus1-dup19250, Lx9-dup44968, L1Md-F2-dup16974, Lx3B-dup2078, Lx-
dup2338, Lx5-dup1453, MMVL30-int-dup87, RLTR6-Mm-dup18, B1-Mus2-dup5988,
MLT1A0-dup8787

B2-Mm2-dup86522, B1-Mus1-dup21242, B1-Mur4-dup9279, RLTR3-Mm-dup23,
B1-Mur4-dup34716, B1-Mus1-dup21610, B1-Mus1-dup21609, B1F-dup10377,
L1Md-F2-dup23486, B2-Mm2-dup20593

Negative: L1-Mur1-dup3007, L1-Mur1-dup3008, B1-Mm-dup9547, ORR1D1-dup7350,
PB1D10-dup14785, MTC-dup12054, B2-Mm2-dup4040, B3-dup122674, RMER12C-dup1049,
L1MA5-dup3764

B1-Mus2-dup3098, B2-Mm2-dup73270, L1MEd-dup1349, L1Md-F3-dup12619,
L1Md-F2-dup50508, L3-dup7855, Tigger7-dup298, RSINE1-dup29247, B1-Mus2-dup57396,
ID4--dup10814

B1-Mus2-dup69004, L1MB7-dup6761, L1Md-F3-dup15875, RMER15-dup10472,
L1Md-F2-dup26830, PB1D9-dup28225, B4-dup52848, Charlie4z-dup46, B4A-dup88275,
L1-Mus1-dup3608



```
[41]: objTE_stellarscope <- FindNeighbors(objTE_stellarscope, dims = 1:10, k.param = 20)
      objTE_stellarscope <- FindClusters(objTE_stellarscope, resolution = 1.2)
      objTE_stellarscope <- RunUMAP(objTE_stellarscope, dims = 1:10)
      DimPlot(objTE_stellarscope, reduction = "umap")
```

Computing nearest neighbor graph

Computing SNN

Modularity Optimizer version 1.3.0 by Ludo Waltman and Nees Jan van Eck

Number of nodes: 4787

Number of edges: 163026

Running Louvain algorithm...

Maximum modularity in 10 random starts: 0.8021

Number of communities: 15

Elapsed time: 0 seconds

14:03:26 UMAP embedding parameters a = 0.9922 b = 1.112

14:03:26 Read 4787 rows and found 10 numeric columns

14:03:26 Using Annoy for neighbor search, n_neighbors = 30

14:03:26 Building Annoy index with metric = cosine, n_trees = 50

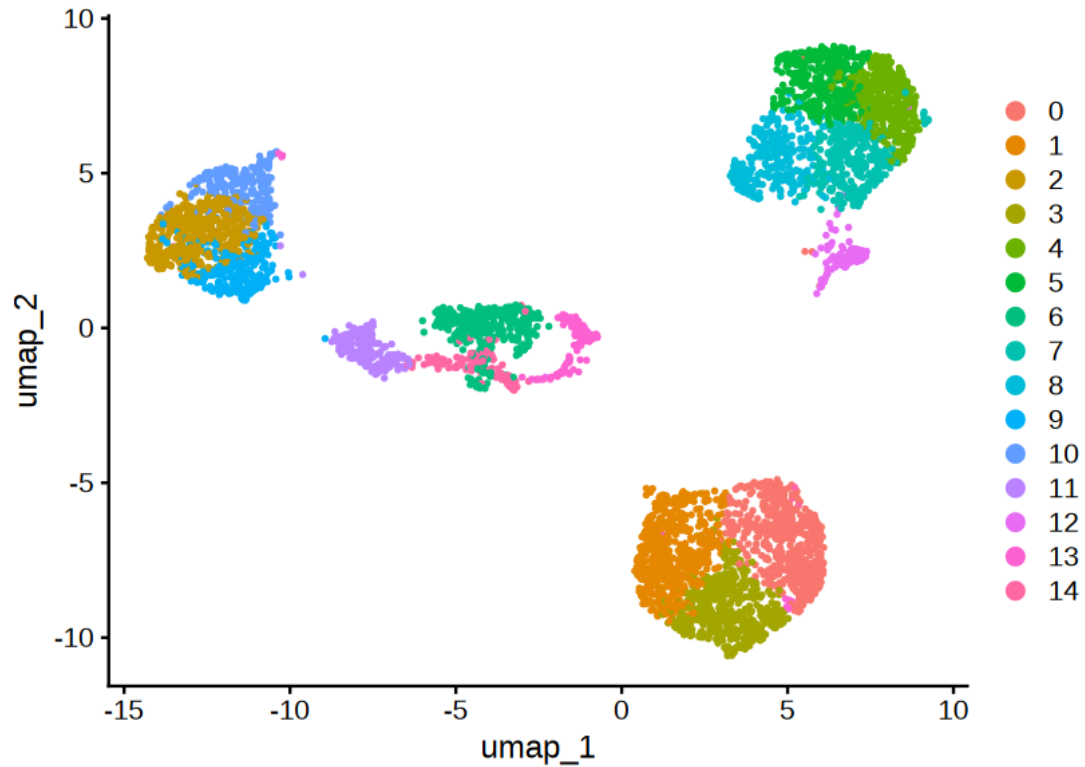
0% 10 20 30 40 50 60 70 80 90 100%

[----|----|----|----|----|----|----|----|----|

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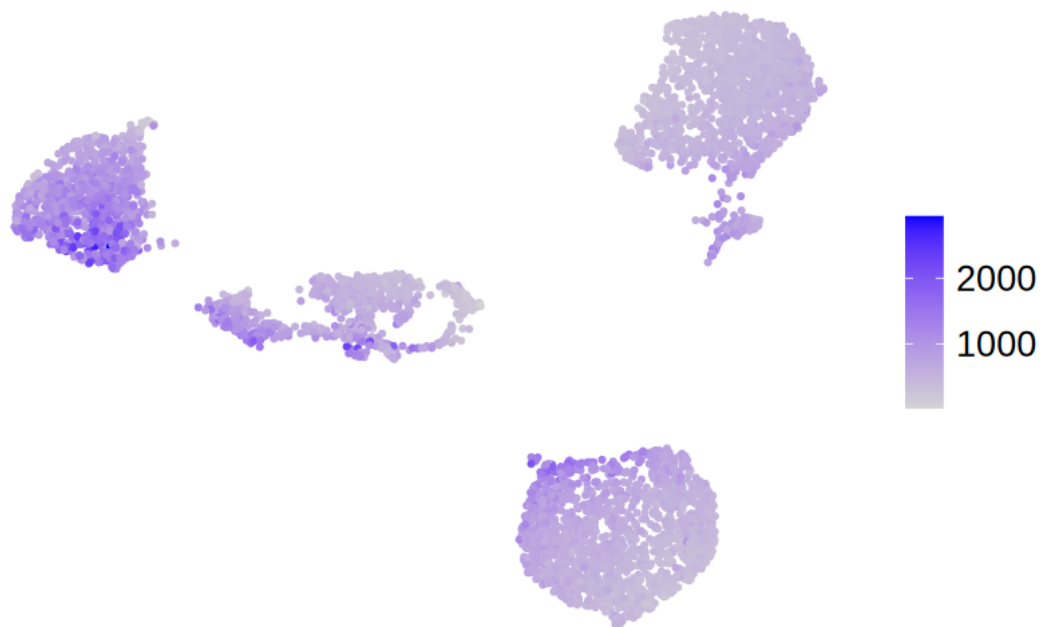
```
14:03:26 Writing NN index file to temp file /tmp/Rtmpkt5V5P/file3ac0d210f7a991
14:03:26 Searching Annoy index using 1 thread, search_k = 3000
14:03:28 Annoy recall = 100%
14:03:30 Commencing smooth kNN distance calibration using 1 thread
with target n_neighbors = 30
```

14:03:33 Initializing from normalized Laplacian + noise (using RSpectra)
14:03:33 Commencing optimization for 500 epochs, with 193092 positive edges
14:03:33 Using rng type: pcg
14:03:40 Optimization finished



```
[42]: FeaturePlot(objTE_stellarscope, reduction = "umap", features = "nCount_TE", pt.  
size = 0.5) +  
theme_void() +  
theme(text=element_text(size=20))
```

nCount_TE



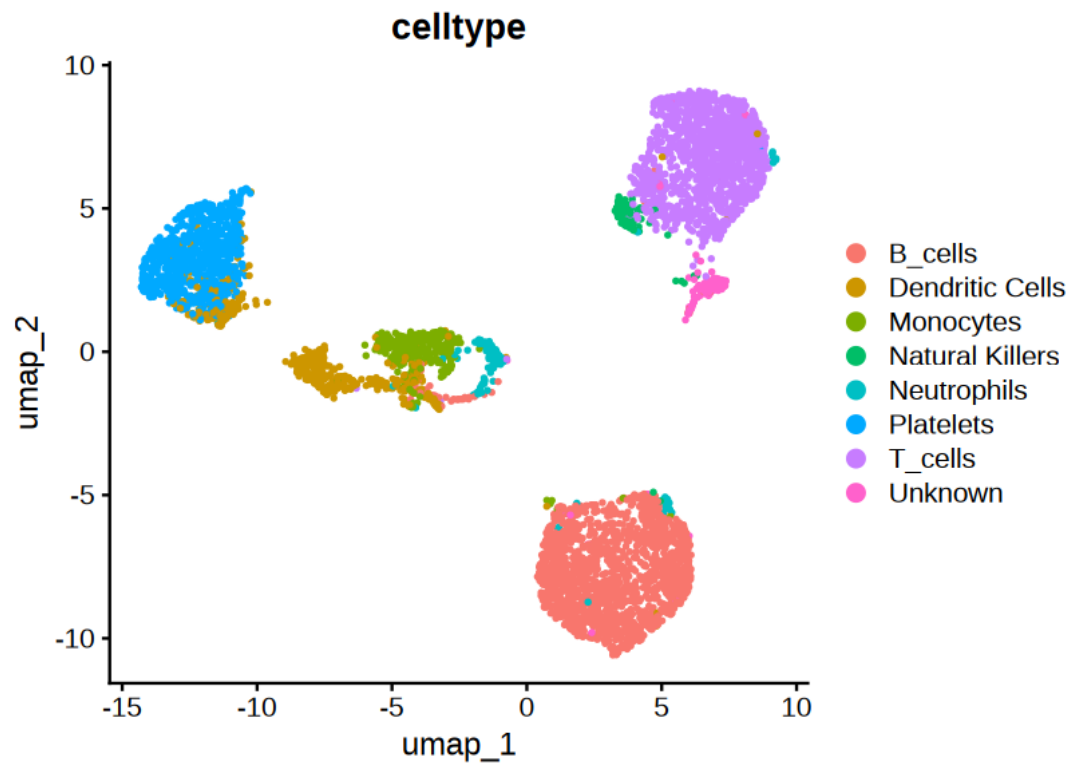
```
[43]: objTE_stellarscope$celltype <-  
      ↪ celltypes$celltype[match(Cells(objTE_stellarscope), celltypes$barcode)]
```

```
[44]: table(objTE_stellarscope$celltype)  
      sum(is.na(objTE_stellarscope$celltype))
```

B_cells	Dendritic Cells	Monocytes	Natural Killers	Neutrophils
1601	438	307	90	113
Platelets	T_cells	Unknown		
828	1294	116		

0

```
[45]: DimPlot(objTE_stellarscope, reduction = "umap", group.by="celltype")
```



```
[46]: saveRDS(objTE_stellarscope, paste0("data_", dataset_id, "/stellarscope_",
      ↪ dataset_id, "_", "seuratObj.RDS"))
```

```
[ ]:
```